EECS 581 - Group 8

Cooper Wright, John Rader, Evan Noeth, Sina Asheghalishahi, Emily Farley, Peter Barybin, Andrew Huang

Student Dashboard

Project Synopsis:

Develop a smart display integrating Canvas assignments, class schedules, weather forecasts, and real-time notifications through APIs and a user-friendly GUI.

Architecture Description (500–1000 words):
This project involves pulling data from various APIs (such as Canvas and weather data), storing a user's class schedule, a centralized GUI to organize the retrieved data, and a hardware screen for displaying the GUI. We will break down the project architecture into 6 sections below for clear organization and include 1 additional section for the API to GUI connection.

Canvas API: The canvas module retrieves assignments from the Canvas REST API. It uses the device's Wi-Fi connection and includes user authentication so the data from one specific student can be accessed. From the retrieved assignments, the module searches for those due on the current day and returns them as a list formatted for the LCD display program. That list in turn feeds the dashboard GUI, which displays the Canvas data and updates when new data arrives. The module also tracks new announcements and grades so the GUI can show an indicator with the number of new tracked items.

Weather API: The weather module connects to a weather data API over the device's Wi-Fi connection and retrieves the weather forecast for the current day. Based on that forecast, it generates a short message for the LCD. The forecast and message are passed to the dashboard GUI, which displays the weather data and updates when new data arrives.

Class Schedule logic: This section will handle the input logic for a class schedule since Canvas itself doesn't internally have the users actual class schedule, just their assignments' schedules. So on the computer running the project, we will make a small GUI for a user to input their schedule where it will then be saved in a separate file for reuse.

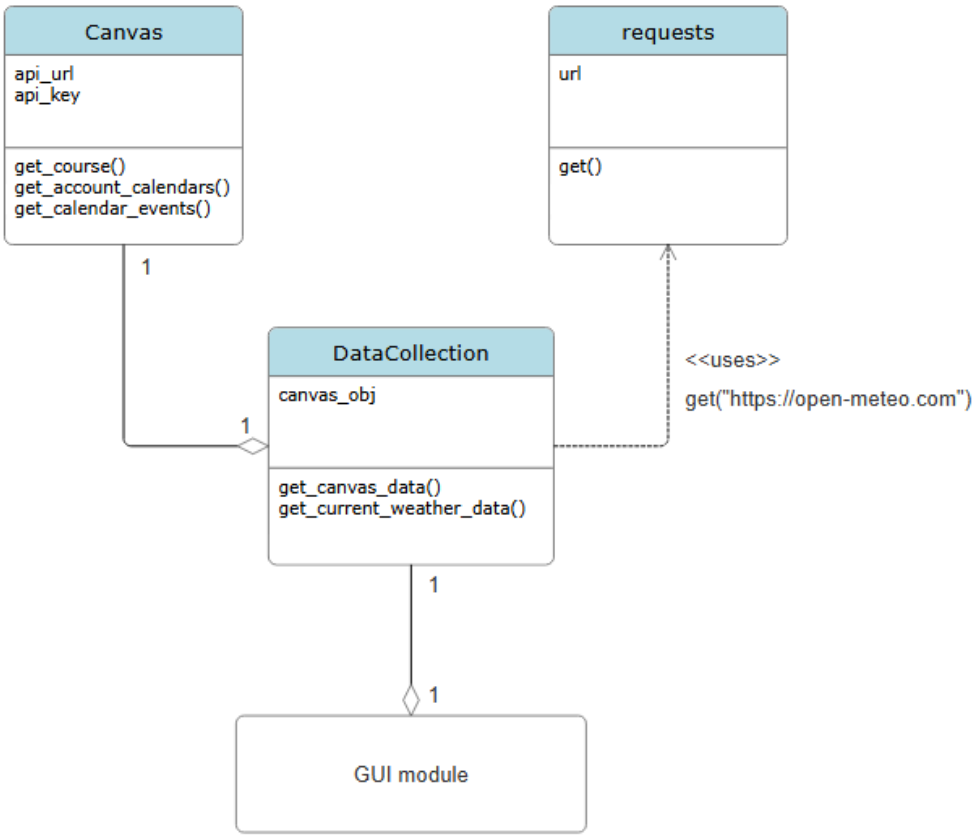
Dashboard GUI: This section will congregate all of the data gathered from the APIs/schedule into a detailed GUI containing sections for each of the APIs and the class schedule. Things that will be displayed include: A schedule for the day, notes on weather for the day, and assignments due for the day. There will also be indicators for new messages/announcements and new grades.

Hardware Level:
The project will be run locally on an ESP32-S3 microcontroller. The exact board that our group is planning on using is unknown, but it will likely be compact, like the Seeed Studio XIAO ESP32-S3. Power for the microcontroller will be supplied from a small 1100 mAh 3.7 V LiPo battery. The wiring will be done by manually soldering connections between the ESP32-S3 board and the lipo and display. However, for the purposes of testing, header pins and DuPont wires will be used instead. The display will be a 240x280, 1.69 in. LCD, and also makes use of an SPI wiring connection.
The code for the microcontroller will be built and flashed using PlatformIO.

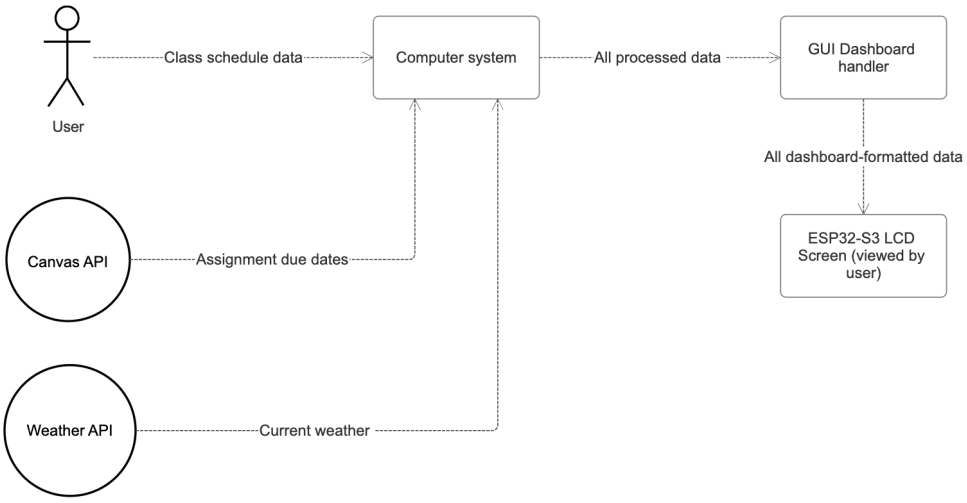
API to GUI connection: The APIs will be called in their own files and data will be formatted and collectable through functions (like get_canvas_data()) which will be called in the GUI file where it will be used and placed appropriately

3 distinct UML diagrams:

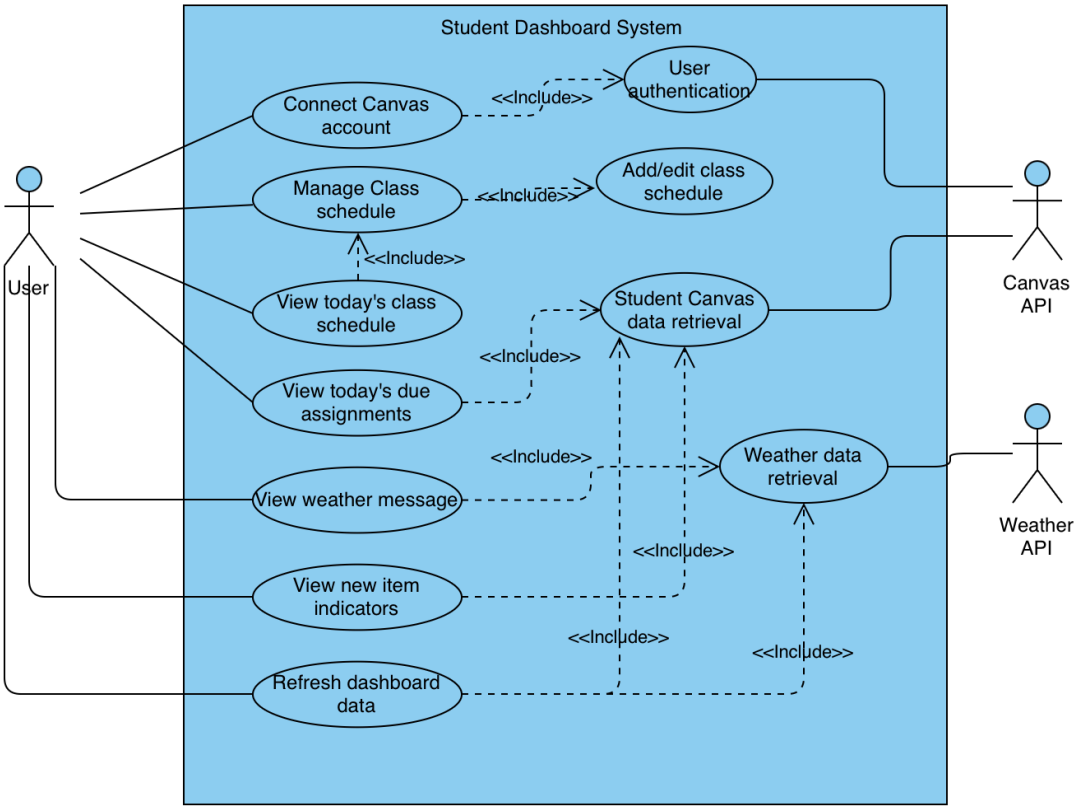
Diagram on API calls



Information Flow Diagram



Use Case Diagram


Student Dashboard System
User
Connect Canvas account
Manage Class schedule
View today's class schedule
View today's due assignments
View weather message
View new item indicators
Refresh dashboard data
User authentication
Add/edit class schedule
Student Canvas data retrieval
Weather data retrieval
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Canvas API
Weather API